Which screens for machine-written scholarly text survive an adaptive author?

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Abstract

Publishers and research-integrity offices screen manuscripts for machine authorship, but detectors are usually evaluated on text that no author has tried to disguise. We audit nine screens: two transparent count-based screens built from seven linguistic events, six published detectors (log-likelihood, log-rank, Binoculars, Fast-DetectGPT, a fine-tuned RoBERTa classifier and a TF-IDF classifier) and a strong zero-shot language-model judge, on 474 human papers and machine papers from three generators, under six attacks. No detector is uniformly robust. Fast-DetectGPT and supervised classifiers are nearly perfect on two generators but fall to AUC 0.61 and 0.60 on the third when half of the sentences are replaced by human text. Probability-based scores invert under perturbation, because disguised text is less probable than genuine human text. Removing structural artifacts leaves supervised accuracy intact within a generator but reduces it across generators. The count screen is weaker than Fast-DetectGPT and the supervised classifiers on every generator, though stronger than likelihood-based scores on Claude, and its worst case under attack (AUC 0.58 to 0.68) is similar to theirs only for Claude. Its value is transparency, not accuracy.

Large language models make fluent scientific prose cheap, and publishers, preprint servers and integrity offices need screens that tell machine-written papers from human ones. Published detectors report accuracy on unmodified output of the generators they were built for[1,2]. A deployed screen faces two further problems: new generators appear continually, and an author who knows how papers are screened can edit the text. Both problems are under-measured. Cross-generator transfer is rarely reported, adaptive authors are usually limited to paraphrasing, and screens that use hand-countable features, which an editor can inspect and explain, are seldom compared with published detectors.

Prior evaluations point to several weaknesses. Detectors lose accuracy under paraphrasing[3,4], on generators and domains absent from their training data[5,6], and can be biased against non-native writers[7]. Watermarking[8] avoids detection altogether but needs the cooperation of every generator and is itself vulnerable to paraphrasing[3]. The stakes are not hypothetical: corpus-level estimates suggest that a growing share of scientific papers already show signs of LLM modification[9]. Lightweight classifiers on hand-designed features have been proposed as a cheap first line of

defence[10], but they are rarely compared with probability-based and supervised detectors under attack. Existing benchmarks cover paraphrasing and unseen generators; they do not combine attacks aimed at the detector's own score (sentence mixing, noise) with a control for structural artifacts in the corpora, which is the gap this audit addresses.

We audit screens under a common protocol. The screens are a transparent count screen, whose seven features are counts of parentheses, square brackets, semicolons, quotation marks, discourse markers, passives and nominalizations, with and without scale-dependent clustering features; six published detectors; and a zero-shot language-model judge. We test each against three generators (GPT-4o, Claude Sonnet 5.5 and Llama 3.3 70B) and six attacks that range from an edit informed by the count screen's features to attacks aimed at probability-based detectors: replacing half of a paper's sentences with human sentences, injecting typographical noise, and rewriting twice. We ask which detectors lose accuracy, which generalize across generators, and whether strong supervised baselines rely on structural artifacts of how the texts were obtained.

This is an independent re-implementation: the original data and code of an earlier version of this work were not available, so the corpus, the generation, the screens and all attacks were rebuilt, and the numbers below differ from those of any earlier version. We report negative results as they came.

Results

Without an adversary, supervised classifiers and Fast-DetectGPT beat the count screen, and Claude is hard for every method

Table 1 gives the AUC of each method against 200 unattacked human papers, using 100 machine papers per generator. Fast-DetectGPT is near perfect on GPT-4o and Llama papers and reaches 0.94 on Claude papers. RoBERTa and TF-IDF classifiers trained on the other two generators exceed 0.93 on every generator. Likelihood-based scores behave much worse on Claude (log-likelihood 0.56, Binoculars 0.67), and the Gemini judge reaches 0.82 on Claude against 0.98 on Llama. The count screen, trained without the target generator, reaches 0.78 to 0.93. Scale-dependent clustering features improve it on all three unseen generators (AUC 0.853 to 0.886 for GPT-4o, 0.781 to 0.810 for Claude and 0.792 to 0.861 for Llama on the full corpus), by +0.03 to +0.07, but the interval for Claude includes zero. A count screen trained on human papers alone does not work (AUC 0.620, 0.471 and 0.663). Claude is the hardest generator for every method.

Table 1 | Detection without attack. AUC against 200 unattacked human papers, 100 machine papers per generator. Count screens are trained on human papers and the other two generators; TF-IDF and RoBERTa likewise (Methods).

Detector	Claude	GPT-4o	Llama	Cost per document
Count screen (rates, dispersion)	0.78	0.85	0.84	ms, CPU
Count screen (+ clustering)	0.80	0.89	0.93	ms, CPU
Log-likelihood	0.56	0.93	0.99	GPU, 7B model
Log-rank	0.59	0.95	1.00	GPU, 7B model
Binoculars	0.67	0.83	0.97	GPU, two 7B models
Fast-DetectGPT	0.94	1.00	1.00	GPU, two 7B models
TF-IDF logistic	0.93	0.98	1.00	ms, CPU (needs labels)
Fine-tuned RoBERTa	0.96	0.99	0.99	GPU fine-tuning (needs labels)
Gemini 2.5 Pro judge	0.82	0.95	0.98	US$0.017, 23 s

No detector is robust to every attack

Figure 1 and Table 2 give the AUC of every method under every attack. The picture differs by method rather than by attack. Fast-DetectGPT is untouched by the feature edit, the LLM rewrite and the hybrid on all three generators, but sentence mixing lowers it for Claude to 0.61. The fine-tuned classifier behaves similarly (Claude, mixing: 0.60). The count screen is hurt by the attack designed against it (for example Llama, feature edit: 0.84 to 0.64) and, for GPT-4o papers, by the hybrid (0.85 to 0.51). Sentence mixing is the strongest attack on most methods. Under mixing the count screen reaches 0.58 to 0.68, which is similar to Fast-DetectGPT, RoBERTa and TF-IDF on Claude papers (0.61, 0.60 and 0.64) but far below them on GPT-4o and Llama papers (0.74 to 0.98). A rewrite alone leaves the count screen unchanged or better (AUC 0.82 to 0.92). TF-IDF is the only supervised or published detector that a rewrite lowers substantially for one generator: for GPT-4o papers its AUC falls from 0.98 to 0.83 after one rewrite and 0.72 after two.

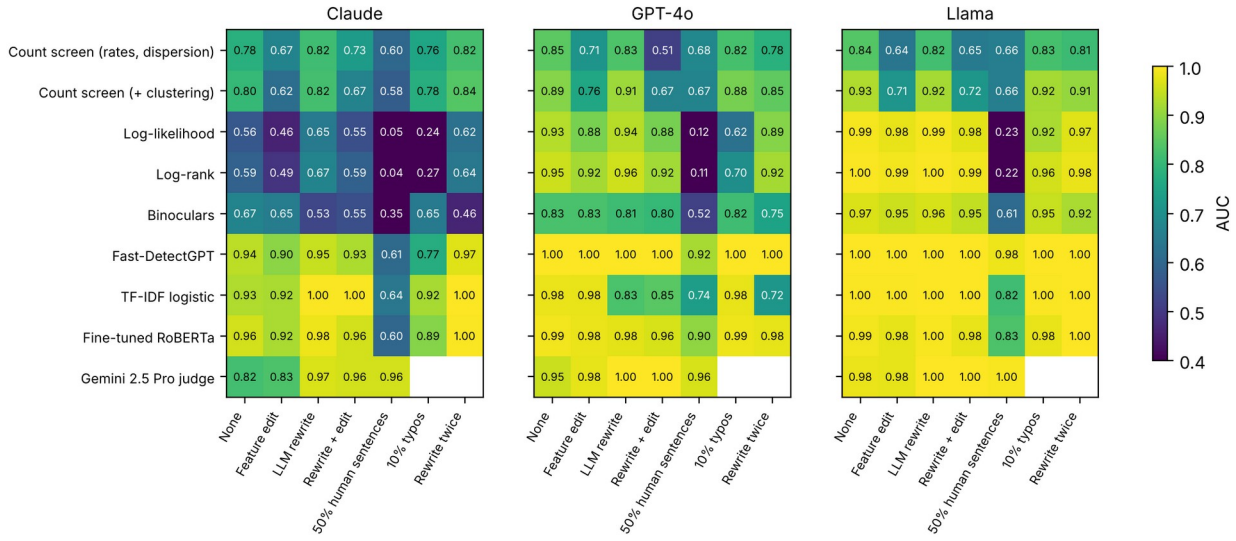


Fig. 1 | AUC of nine screens under seven conditions. Each panel is one generator; negatives are 200 unattacked human papers. Attacked sets contain 100 papers (the Gemini judge was run on four attacks and, for mixing, stopped at a spending cap after 99, 100 and 64 papers for Claude, GPT-4o and Llama).

Table 2 | Worst case under attack. AUC without attack, followed by the lowest AUC over six attacks (in brackets, the attack). For the judge the attacks are the feature edit, the rewrite, the hybrid and sentence mixing.

Detector	Claude	GPT-4o	Llama
Count screen (rates, dispersion)	0.78 → 0.60 (50% human sentences)	0.85 → 0.51 (rewrite + edit)	0.84 → 0.64 (feature edit)
Count screen (+ clustering)	0.80 → 0.58 (50% human sentences)	0.89 → 0.67 (rewrite + edit)	0.93 → 0.66 (50% human sentences)
Log-likelihood	0.56 → 0.05 (50% human sentences)	0.93 → 0.12 (50% human sentences)	0.99 → 0.23 (50% human sentences)
Log-rank	0.59 → 0.04 (50% human sentences)	0.95 → 0.11 (50% human sentences)	1.00 → 0.22 (50% human sentences)
Binoculars	0.67 → 0.35 (50% human sentences)	0.83 → 0.52 (50% human sentences)	0.97 → 0.61 (50% human sentences)
Fast-DetectGPT	0.94 → 0.61 (50% human sentences)	1.00 → 0.92 (50% human sentences)	1.00 → 0.98 (50% human sentences)
TF-IDF logistic	0.93 → 0.64 (50% human sentences)	0.98 → 0.72 (rewrite twice)	1.00 → 0.82 (50% human sentences)
Fine-tuned RoBERTa	0.96 → 0.60 (50% human sentences)	0.99 → 0.90 (50% human sentences)	0.99 → 0.83 (50% human sentences)
Gemini 2.5 Pro judge	0.82 → 0.83 (feature edit)	0.95 → 0.96 (50% human sentences)	0.98 → 0.98 (feature edit)

Edit strength acts gradually on the count screen and barely on the other detectors

The feature edit moves counts a share e of the way toward the human mean. The count screen declines steadily with e (AUC for the baseline screen, e = 0, 0.25, 0.5, 0.75, 1: Claude 0.78/0.75/0.71/0.70/0.67; GPT-4o 0.85/0.82/0.79/0.76/0.71; Llama 0.84/0.79/0.73/0.71/0.64), a loss roughly proportional to effort that reaches 0.11 to 0.21 AUC at full effort. Fast-DetectGPT

and the classifiers change by less than 0.06 at full effort. This is what one expects when an attack targets the features of one detector and not another's: the edit moves counts that the count screen uses, but leaves the word choices and the probability structure on which the other detectors rely. A hybrid that rewrites first and then edits (Supplementary Table 1 to 3) is stronger than the edit alone against the count screen for GPT-4o papers and no stronger for the others.

Probability scores invert under perturbation

Log-likelihood and entropy fall below chance for perturbed text: for Claude papers with half of the sentences replaced by human sentences, the AUC of log-likelihood is 0.05 and that of entropy 0.03. This is not a scoring error. The perturbed text is less probable than genuine human text (mean log-likelihood per token -2.72 against -2.04) because spliced sentences break coherence and typos lower probability, so a rule that calls high-probability text machine-written labels perturbed text human. A two-sided score, the absolute deviation from the human median, recovers detection (log-likelihood 0.90, entropy 0.94; for GPT-4o papers 0.12 against 0.75). The deviation threshold was set on the same human papers, so these are optimistic, but it shows that an author who wants to pass a one-sided probability detector can overshoot, and that a defender can look for text that is too improbable as well as too probable. Fast-DetectGPT, which normalizes by the curvature of the model's own distribution, does not invert.

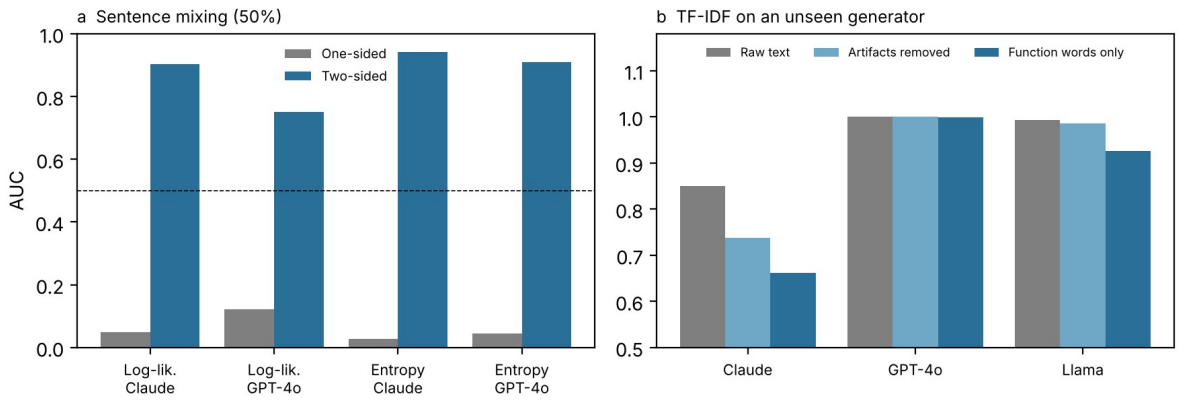


Fig. 2 | Two controls. a, AUC of one-sided and two-sided probability scores for Claude and GPT-4o papers with half of the sentences replaced by human sentences (two-sided: absolute deviation from the human median). **b**, AUC of the TF-IDF classifier on an unseen generator for three versions of the input text.

Strong supervised accuracy rests partly on artifacts and does not transfer fully

The TF-IDF classifier separates human from machine text almost perfectly (Table 1), and the human and machine texts differ in structure: human text is extracted from published PDFs and

XML and contains headers, 'Fig.' and 'Table' labels, 'et al.', years and affiliations, whereas machine papers contain none. We compared three versions of the classifier (Table 3). Removing the text before the abstract, digits and a lexicon of such tokens leaves accuracy intact when the generator is in the training set, and function words alone still give AUC 1.00. Function-word differences are real stylistic signals ('thus', 'hence', 'indeed' for humans; 'across', 'often', 'such', 'beyond' for machines). The fall appears on an unseen generator: for Claude the AUC drops from 0.85 to 0.74 when artifacts are removed and to 0.66 with function words only, and for Llama from 0.99 to 0.93. Supervised detectors therefore perform best exactly where a deployed screen cannot count on being: when training data from the generator in question exist.

Table 3 | Does the TF-IDF classifier depend on structural artifacts? AUC of a logistic classifier on word and bigram TF-IDF of the first 2,000 words. 'Same generator' is five-fold cross-validation grouped by source paper; 'unseen generator' trains on the other two generators and half of the human papers and tests on the remaining human papers and the held-out generator.

Protocol	Generator	Raw text	Artifacts removed	Function words only
Same generator in training	Claude	1.00	1.00	1.00
Same generator in training	GPT-4o	1.00	1.00	1.00
Same generator in training	Llama	1.00	1.00	1.00
Unseen generator	Claude	0.85	0.74	0.66
Unseen generator	GPT-4o	1.00	1.00	1.00
Unseen generator	Llama	0.99	0.99	0.93

The judge is strong, notices damaged text and has one blind spot

The Gemini 2.5 Pro judge flags 1% of human papers. Without attack its AUC is 0.98 for Llama papers and 0.82 for Claude papers; the GPT-4o judge is worse, at 0.75, 0.64 and 0.92 for GPT-4o, Claude and Llama papers on the full judged sample, and flags none of the Claude papers. Rewriting raises the Gemini judge's AUC on Claude papers to 0.97, and so does mixing in human sentences (0.96), probably because spliced text reads as incoherent. The judge is therefore robust to the attacks here, apart from a weak starting point for Claude papers when text is only edited (AUC 0.83), and it pays for this: US$0.017 and 23 s per excerpt, against milliseconds for the count screen, and its robustness comes from seeing incoherent text that a genuine author would not produce, not from recognizing the generator.

Discussion

Three points follow. First, robustness is a property of the pair of detector and attack, and no detector in this audit dominates. A deployed screen should therefore combine detectors whose failures are different: a probability-based detector (Fast-DetectGPT) is near perfect against edits and rewrites, a supervised classifier adds style, and a judge catches damaged text. Second, probability-based scores are one-sided, and an adaptive author can push text beyond the human range; a two-sided check costs little. Third, the strongest baseline in a benchmark can owe part of its performance to how the corpora were assembled. We found artifact dependence in cross-generator transfer, and any comparison of a new detector with a supervised baseline should include such a control.

These results suggest a short checklist for evaluating a machine-text detector: report accuracy on a generator that played no part in training; include at least one attack that targets the detector's own score and not only paraphrasing; check whether a high score survives removal of structural artifacts; and, for probability-based scores, test a two-sided rule. Our data show each item changing a conclusion: Claude papers separate the detectors that look alike on other generators, sentence mixing separates detectors that look alike under paraphrasing, and artifact removal separates TF-IDF from the other supervised detector on unseen generators.

Claude papers are the hardest for every method, and the reason is not established by our data. Candidate explanations are that the model's output is less predictable under the open model used for probability scores, that its prose contains more of the features human authors use, or that it was generated with a reasoning mode that changes style. We did not test them, and a ranking of generators by difficulty is probably not stable across prompts and genres. What the audit does show is that a screen validated on one or two generators can fail on a third in ways that headline accuracy hides, which is the practical case for a held-out generator in every evaluation.

The judge deserves a separate caution. It reads an excerpt and gives a probability, and part of its strength under mixing comes from incoherence that a careful author would not produce; an attacker who mixes sentences from a single related source, or who repairs coherence with a rewrite after mixing, might not be caught. The GPT-4o judge also evaluates text from its own family, and its verdicts on GPT-4o papers involve self-recognition. Neither issue was tested here.

The transparent count screen did not match published detectors, so we do not claim it as a better detector. Its case rests on explanation and on cost: every feature can be shown to an author or a

committee, and scoring a document takes milliseconds. Clustering features helped on unseen generators. Whether explainability is worth the lost accuracy is a policy question that this audit does not answer.

The limits are substantial. The study covers three generators, one genre (English-language scientific papers), 100 attacked papers per generator and cell, so AUC differences below about 0.07 are within sampling error (Methods), and 200 human negatives. The Gemini judge was run on four attacks and on mixing for 263 of the 300 papers because of a spending cap. Machine papers were generated in four requests from the title and abstract of a real paper, human papers come from two extraction pipelines, and both facts create artifacts that we only partly controlled. The attackers are simple: none optimizes against a detector's gradient, uses a paraphraser trained against detectors or edits at the token level with a surrogate. Zero-shot detectors used one open model pair (Falcon-7B and its instruct version). We did not test light human editing of machine text or the use of detectors on mixed human–machine documents, and the economic questions of base rates and review costs are not addressed here.

Methods

Documents

We sampled 500 human papers at random from 2015 to 2019, before large language models were widely used for scientific writing: 330 from the open-access subset of PubMed Central (article XML body text) and 170 from arXiv (PDFs converted to text, reference lists removed, broken lines rejoined). We kept papers with at least 1,500 words, a title and an abstract. Features need four 500-token windows, so papers with fewer than 2,000 tokens after cleaning were dropped from the analysis, leaving 474 human papers (309 PubMed Central, 165 arXiv; median length 4,299 words before truncation). For each sampled paper a generator wrote a paper on the same topic from the original title and abstract, with one of three prompts of increasing detail (temperature 0.7), through a commercial gateway. Each paper was written in four requests (introduction, methods, results, discussion with conclusion) and concatenated. Every document was truncated to its first 2,000 whitespace-delimited tokens (Table 4).

Table 4 | Corpus. Documents per source before truncation.

Source	Documents	Median words	10th–90th percentile
Human (PubMed Central, arXiv)	474 analysed (500 collected)	4,299	2,294–8,876

GPT-4o	245	3,639	3,219–4,039
Claude Sonnet 5.5	151	3,476	3,252–3,684
Llama 3.3 70B Instruct	202	3,002	2,595–3,813

Human length statistics are for all 500 collected papers; machine papers are those that passed the feature filter. Claude papers were capped to limit cost.

Count screens

In each of four consecutive 500-token windows we counted seven events: parentheses, square brackets, semicolons, paired quotation marks, fourteen discourse markers, passive constructions and nominalizations (the last two by approximate pattern matching). Features are the mean count per window (rate) and the dispersion index, 14 in total for the baseline screen; the clustering variant adds the mean log Fano factor across window sizes of 50, 100, 200 and 400 tokens and its slope against log window size for each event, for 21 features per document. Both are logistic regressions on standardized features; dispersion indices follow standard count-data practice[11].

Published detectors and judge

Log-likelihood, log-rank, Binoculars[12] and Fast-DetectGPT[2] were computed with Falcon-7B and Falcon-7B-instruct on the first 1,024 tokens of each document. The TF-IDF classifier uses word unigrams and bigrams (at most 100,000 features, logistic regression, C = 3). The RoBERTa classifier is roberta-base fine-tuned for two epochs (maximum length 512, learning rate 2e-5). For each target generator the supervised classifiers were trained on human papers outside the evaluation sample and papers of the other two generators, excluding Llama papers as training data when Llama was the attacker and excluding papers sharing a source with the evaluation sample. The judge prompt asks Gemini 2.5 Pro (temperature 0, up to 8,000 output tokens; zero-shot judging as in [13]) for the probability from 0 to 100 that an excerpt was written by an AI model; excerpts were the first 2,000 tokens with markdown removed.

Attacks and evaluation

Attacks were applied to 100 papers per generator. The feature edit moves counts of five events toward the human mean in two of four windows. The LLM rewrite asks Llama 3.3 70B to rewrite each paper so that it reads as if written by an experienced researcher; the hybrid applies the edit to the rewrite; the second rewrite repeats the rewrite. Sentence mixing replaces 50% of the sentences with sentences from human papers of the same field (not among the negatives); typo noise gives each word of at least four letters a random swap, deletion or repetition of a letter with probability 10%. The count screens, TF-IDF and RoBERTa were never trained on attacked

text. All AUCs compare 200 unattacked human papers with the attacked machine papers of one generator. The artifact control uses five-fold cross-validation grouped by source paper and a held-out-generator protocol.

Statistics

Each cell of Table 2 and Fig. 1 compares 100 attacked machine papers with 200 unattacked human papers. The standard error of an AUC for these sample sizes[14] is 0.035 at AUC 0.60, 0.032 at 0.75, 0.022 at 0.90 and 0.012 at 0.97, so 95% intervals have half-widths of about 0.02 to 0.07. We therefore do not interpret differences smaller than about 0.07 between two methods or two conditions. Intervals for the clean count-screen results in Supplementary Tables 4 to 6 are bootstrap intervals over documents with folds grouped by source paper. Per-document scores of the supervised classifiers and the count screens on attacked text were not retained, so no bootstrap intervals are given for those cells.

Reporting

No human participants were involved. All analyses used published papers and machine-generated text. [Confirm with the authors' institution whether any approval is needed.]

Data availability

Machine-generated papers, features, scores and result tables are in the replication package on Zenodo (https://doi.org/10.5281/zenodo.23664508). Human papers are public (PubMed Central open access, arXiv) and are re-collected with the identifier list in the package.

Code availability

All code is in the replication package on Zenodo (https://doi.org/10.5281/zenodo.23664508), released under the MIT licence.

Author contributions

O.L. conceived and designed the study, built the corpus and the analysis pipeline, interpreted the results and wrote the manuscript.

Competing interests

The author declares no competing interests. Generative AI disclosure: the machine-written papers studied here were produced by GPT-4o, Claude Sonnet 5.5 and Llama 3.3 70B Instruct,

and Gemini 2.5 Pro served as a judge. An AI coding assistant (Claude Code) was used to write and run the analysis code and to draft and edit the manuscript text; the author reviewed the results and takes responsibility for the content.

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